HACKATHON CHALLENGE

Energy Consumption Prediction

Iberia Market · Spain

April 2026

AXPO × Databricks Challenge

APPROACH

Clustering the Data

BY REGION

Very **heterogeneous** clusters with variable sizes

⚠ Geographical but inconsistent

BY BEHAVIOUR

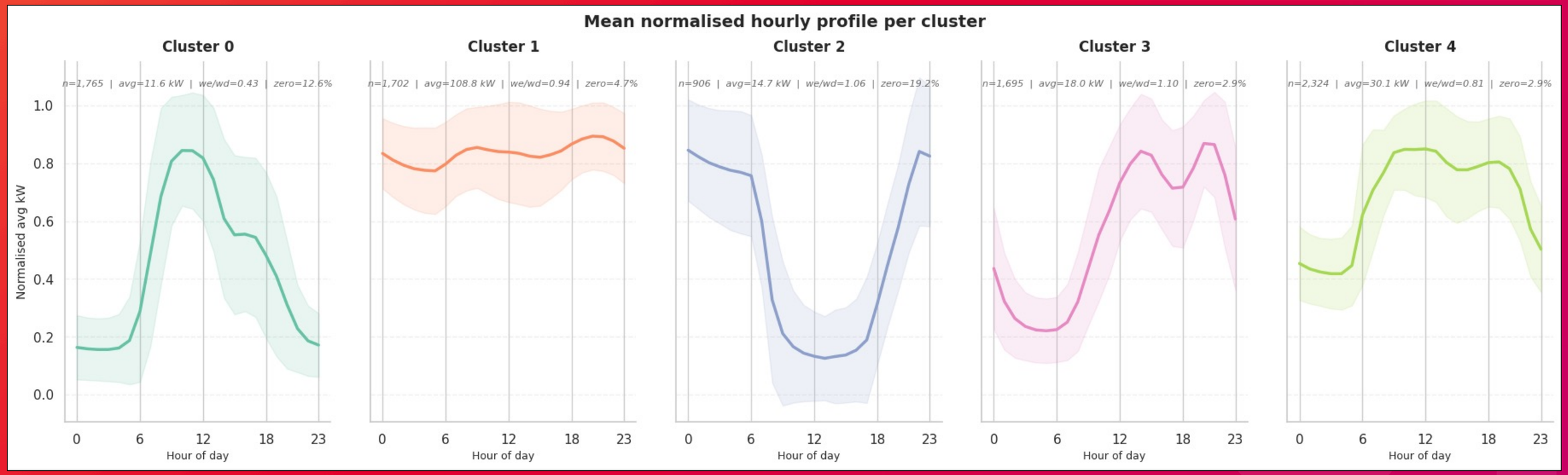
Very **representative** clusters but no geographical consistency

⚠ Behavioural but not spatial

EVEN BETTER

Combine **both methods** using ensemble techniques

✓ **Best of both worlds**



MODELLING

Model & Features

• CHALLENGE

Give the model an idea about the **general pattern** while capturing **recent changes**

Avoiding **data leakage**

• OUR APPROACH

Statistical features + external data to enrich signal quality

e.g. visitors, last measured value, lags...

Simple models — reliability first given submission constraints

e.g. XGBoost, simple statistical predictions

02

THANK YOU

Questions?

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